E-Bin: The Smart E-Waste ATM.

An AI-Powered Collection System for a Closed-Loop Micro-Economy.



Gemini Notebook

The Micro E-Waste Crisis



The Friction

Lack of convenient recycling facilities leads users to throw small electronics into regular bins.



The Consequence

Improper disposal causes toxic materials to leach, heavily polluting soil and water.

The current recycling infrastructure is fundamentally misaligned with human behavior for small-scale e-waste.

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Engineering a Paradigm Shift in E-Waste Disposal

	Traditional Disposal	The E-Bin Paradigm
User Effort	 High (Requires travel to specific facilities)	 Low (Accessible ATM-style drop-off)
Sorting Mechanism	 Manual and downstream	 AI-powered identification and automatic sorting at the point of insertion
Incentive	 None	 Instant reward points and discount coupons
Outcome	 Toxic pollution	 Closed-loop recycling and recovered materials

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System Architecture & User Journey

Step 1:
User inserts un-sorted e-waste into the E-Bin.



Step 5:
User instantly receives rewards (points/coupons) via the mobile app.

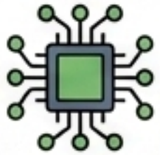


E-Bin Hardware Interface

Step 2:
AI camera scans and identifies the inserted item.



Step 3:
Neural network classifies the specific type of e-waste.



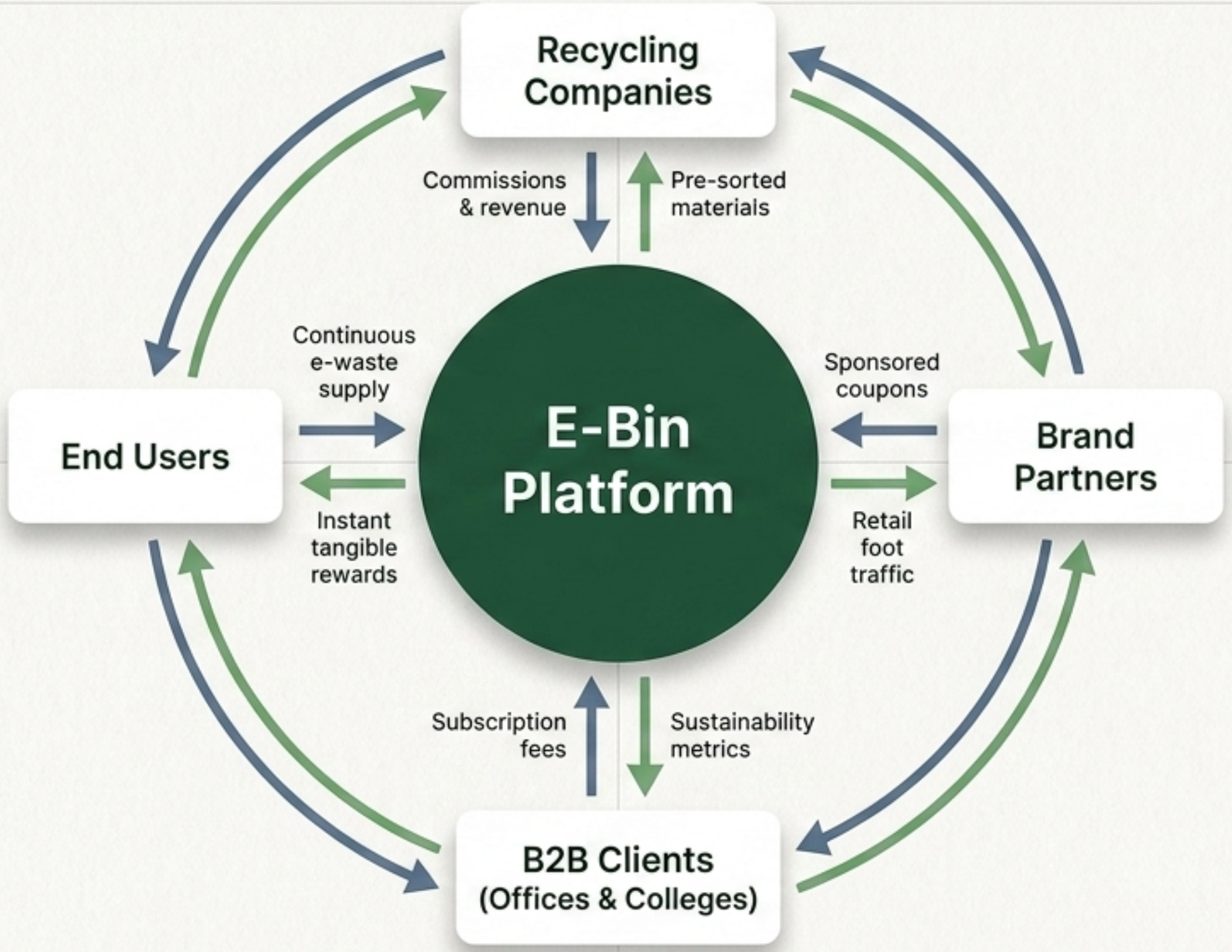
Step 4:
Mechanical routing system automatically stores the item in the correct internal compartment.



Machine Journey

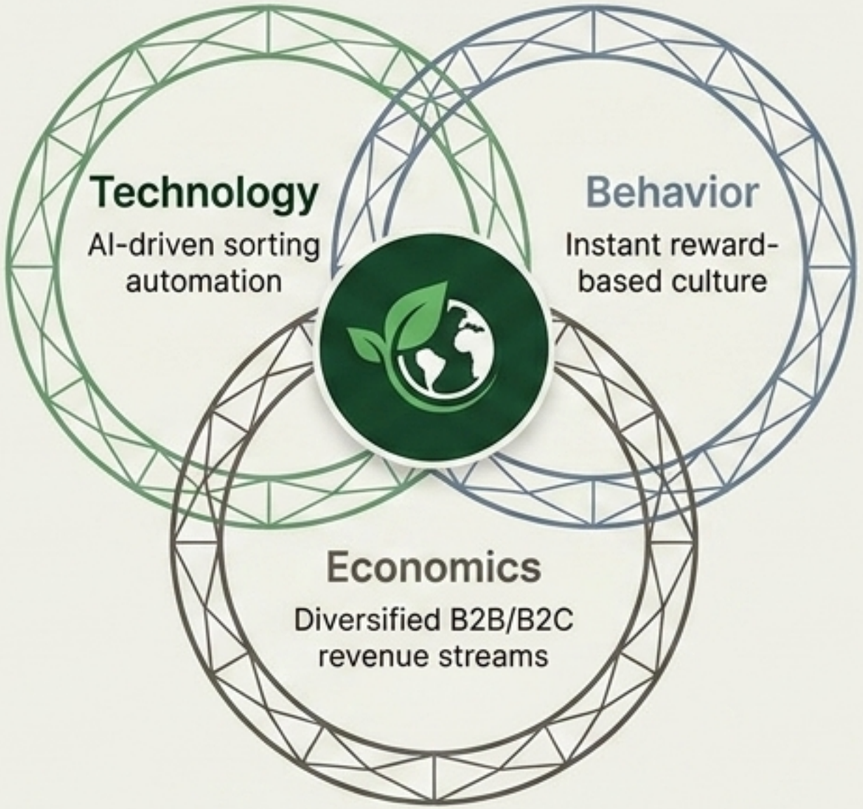
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The E-Bin Economic Ecosystem



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A Self-Sustaining Micro-Economy for a Greener Future



E-Bin makes e-waste recycling simple, rewarding, and accessible while promoting a cleaner and greener environment.

We are not just collecting waste; we are transforming pollution into a profitable, reward-driven recycling culture.

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